Computer Networks — Lecture 3

The OSI and TCP/IP Models: how data travels across a network

1. Why do we need network models?

Sending a message from one computer to another involves many problems at once: turning bits into electrical or radio signals, finding a path through many routers, recovering from lost packets and presenting data to the right application. Network models split this big problem into **layers**. Each layer solves one problem and offers a service to the layer above it. Engineers can change one layer (for example, replace Wi-Fi with fibre) without rewriting the others.

2. The seven layers of the OSI model

#	Layer	Job	Examples	Data unit
7	Application	Services for user programs	HTTP, DNS, SMTP, FTP	Data
6	Presentation	Format, encrypt, compress	TLS, JPEG, UTF-8	Data
5	Session	Open / manage / close sessions	RPC, NetBIOS	Data
4	Transport	End-to-end delivery, ports	TCP, UDP	Segment
3	Network	Logical addressing, routing	IP, ICMP, OSPF	Packet
2	Data Link	Delivery on one link, MAC addresses	Ethernet, Wi-Fi (802.11)	Frame
1	Physical	Bits as signals	Cables, fibre, radio	Bit

Memory trick (top to bottom): *All People Seem To Need Data Processing*.

3. The TCP/IP model

The Internet actually uses the simpler TCP/IP model with four layers. It merges OSI layers 5–7 into one Application layer and layers 1–2 into the Link (Network Access) layer.

- **Application** — HTTP, HTTPS, DNS, SMTP, SSH
- **Transport** — TCP and UDP, identified by port numbers (e.g. 80 for HTTP, 443 for HTTPS, 53 for DNS)
- **Internet** — IP addressing and routing (IPv4, IPv6)
- **Link** — Ethernet, Wi-Fi, and the physical medium

4. Encapsulation

When you send data, each layer adds its own **header** in front of the data from the layer above. This is called encapsulation. The receiver removes the headers in reverse order (decapsulation).

- Application data: GET /index.html
- Transport adds a TCP header (source port, destination port, sequence number) → segment
- Network adds an IP header (source IP, destination IP, TTL) → packet
- Data link adds a frame header with MAC addresses and a trailer with a checksum (FCS) → frame
- Physical layer sends the frame as bits

5. TCP vs UDP

TCP (Transmission Control Protocol) is connection-oriented and reliable: it numbers bytes, acknowledges what was received, retransmits lost segments, keeps order and controls the sending speed (flow and congestion control). It is used for web pages, e-mail and file transfer.

UDP (User Datagram Protocol) is connectionless: no handshake, no acknowledgements, no ordering. Its header is only 8 bytes, so it is fast and has low delay. It is used for video calls, online games, live streaming and DNS queries, where speed matters more than perfect delivery.

6. The TCP three-way handshake

- **SYN**: the client sends a segment with the SYN flag and its initial sequence number x
- **SYN-ACK**: the server replies with SYN and ACK = x + 1 and its own sequence number y
- **ACK**: the client sends ACK = y + 1. The connection is now established

A connection is closed with FIN and ACK segments from both sides (four-way termination).

7. Example: what happens when you open https://example.com

- DNS (application layer, usually over UDP port 53) translates example.com into an IP address
- The browser opens a TCP connection to port 443 using the three-way handshake
- TLS (presentation) negotiates encryption keys
- HTTP sends GET / and the server returns the HTML page
- Routers forward the IP packets hop by hop; switches deliver frames using MAC addresses

8. Key points for the exam

- Name the seven OSI layers in order and give one protocol for each
- Compare the OSI and TCP/IP models
- Explain encapsulation and name the data unit at each layer
- Compare TCP and UDP and give suitable applications
- Describe the three-way handshake with sequence and acknowledgement numbers